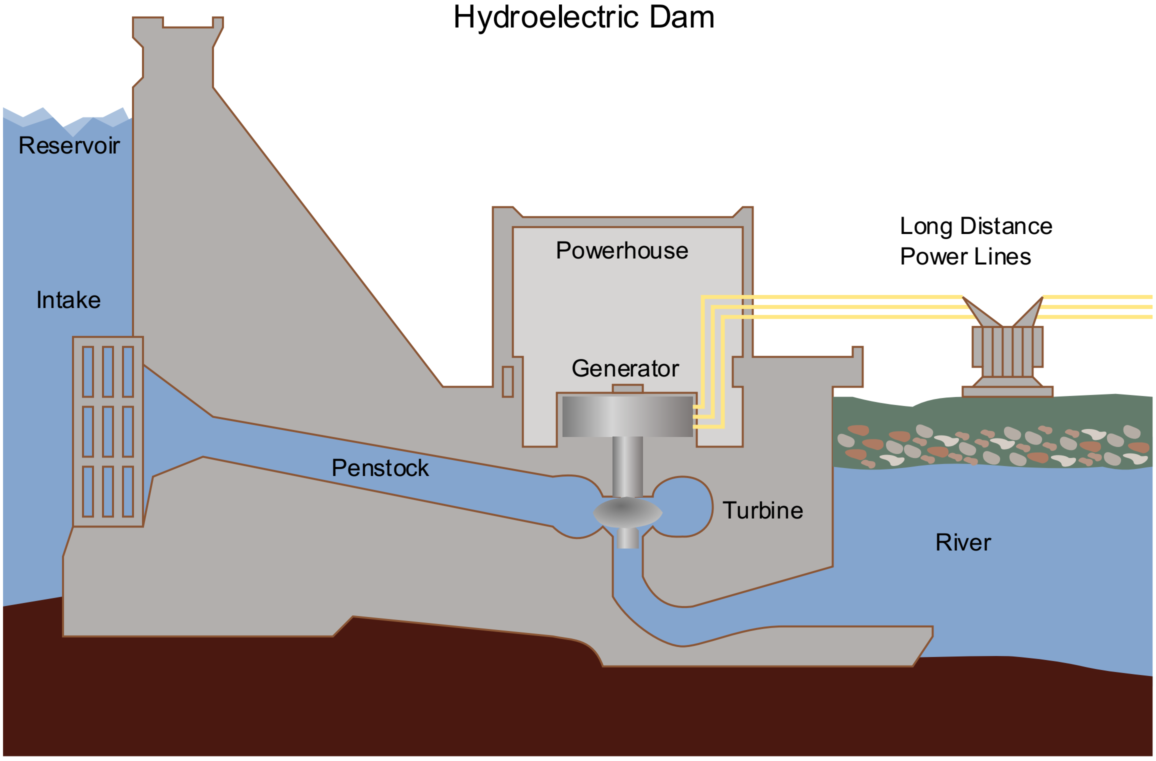
Hydroelectric and wind energy

We are going to learn about two important renewable energy resources.
One is Hydroelectric power and another one is wind energy
First we begin with Hydroelectric Power
Hydro-energy or hydropower is known as one of the traditional renewable energy resources.
Harvesting energy from flowing water or falling water into electricity is known as Hydroelectric energy.
Globally hydroelectricity accounts for more than 18% of all electricity produced and Hydroelectricity is the fourth largest energy resource across the whole world.
Hydro-energy is known as a traditional renewable energy resource. The principle behind the hydroelectric power generation is relatively very simple and has been existing for a significant span of time.
It has been in use for many centuries. Hydro energy was mentioned even in the Greek poems of the 4th century.
If you travel to European countries you can still see large waterwheels. The Romans were the first to utilize the waterwheel. They have used these waterwheels mostly for grinding grains.
On September 30 1882 the world's first hydroelectric power plant began operation on the Fox River in Wisconsin. This is the first example of converting water energy into electricity.



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There are two methodologies in use that utilize water to produce electricity
One is electricity from a hydroelectric dam and another one is electricity from a pumped-storage plant.
Let's see how exactly is electricity generated from the dam?
The principle is simple:
The source of hydroelectric power is water; Thus hydroelectric power stations are usually constructed on a large river.
The dam has massive walls that blocks the flow of a stream or a river.
It allows accumulation of lot of water in the reservoir of the dam. At the bottom of the dam or large reservoir there is an intake from which the water delivers through a special channel called penstock. When an intake opens the nozzle system of penstock forces water to flow through a channel. The water rushes to the hydraulic turbine water get accelerated and then water hitting the turbines blades and causing it to spin. The turbine is connected by a shaft. Rotating turbine causes the shaft to rotate. The rotating shaft is connected to an electrical generator which converts the mechanical energy of the shaft into electrical energy. That is converting waters kinetic energy into mechanical energy subsequently into electricity.

Dams stores a lot of water at a higher level Gravity causes water to fall to rotate turbines that generate electricity. After passing through the turbine the water returned to the river and continued its journey.

Opening underwater gates directly controls the amount of water flowing through the special channel determining the amount of electricity generation from the dams.

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The second methodology to harvest hydropower is through Pumped-storage plants. A pumped-storage plant is very similar to a dam-based hydroelectric power stations but the main difference is that the pumped-storage plant uses two reservoirs at different elevations one will be considerably higher than the other. The surplus electrical energy available during low-demand periods or in off-peak periods will be utilized to pump the water from a lower reservoir to an upper reservoir. When the power consumption is low for example the middle of the night the dam uses this surplus electrical energy to pump the water from lower reservoir to the upper reservoir. When high electrical demand occurs the stored water is released. The released water will rotate the turbines and generate electricity. These kinds of systems increase revenue and overcome shortage of electricity and reduces the electricity bills.

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Also Hydroelectric energy can be harvested even on a small scale with the natural flow of run-of-river; the running water drives one or more turbines and generating electricity

However like every source of energy there are several advantages and disadvantages with hydroelectric energy. Let's discuss the benefits and drawbacks of hydroelectric energy.

Advantages of hydroelectric energy
Hydroelectric energy is renewable.
The cost of hydroelectricity is relatively low.
It is cheaper than thermal or nuclear power.
The efficiency of hydropower systems is very high >80% while thermal power plants have low efficiency as low as 40%.
The average cost of electricity from hydropower power station is very less.
The electricity production of a dam is dependent on how much water stored in the reservoirs that's why the output electricity can be determined based on the need.
Meaning that we can adjust water flow and hence we can control the output electricity. For example when power consumption is low water flow can be adjusted consecutively we can control the output electricity.
Hydropower plants are flexible and they can produce electricity at a constant rate.
Besides hydropower does not consume water.
Hydropower is the most reliable efficient and economical.
Once constructed a hydropower power stations will be operational for at least 40–50 years.
Even though construction of large dams is very expensive the produced energy from hydroelectric dam is virtually free.
The water reserved in the dam can be used for irrigation or for leisure purposes such as recreation and fishing.
Also hydroelectric dams release a negligible amount of greenhouse gases.
Furthermore it creates no pollution and it is a relatively cheap and clean source of energy.

Hydropower also has some disadvantages
Dams are very expensive to build
Hydroelectric dams alter the landscape dramatically

Building of dams disturbs and depletes the natural habitats of fisheries and river ecosystems plants and animals.
Large dams have forced people to relocate making people homeless especially tribals.
Building a large dam will of course causes flood. If flooding happens a large area gets submerged. This causing problems for people and the animals who live there.
Another drawback is finding suitable locations to construct the dams is challenging.
The construction of a dam blocks the natural flow of water such as rivers. It can result in serious disputes between neighbouring countries and also neighbouring states.

Let's discuss about second renewable energy I have mentioned.
That is Wind energy


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Wind energy is actually a by-product of solar energy.
Wind energy is one of the most promising fast-growing renewable energy of the future.
Let's discuss Where does wind come from?
Solar energy causes differences in temperature density and pressure in the atmosphere and this causes the movement of air. That's why wind energy is considered as indirect solar energy.
As early as 4000 - 3500 BC Ancient Egyptians used wind energy to sail their ships.
As the water wheel the wind wheel has been used by humans for a long time for grinding grains and pumping water for irrigation and doing other types of work. Windmills even started appearing around 8th and 9th centuries in the middle east and Western Asia. Then slowly arrived in India China and Europe.

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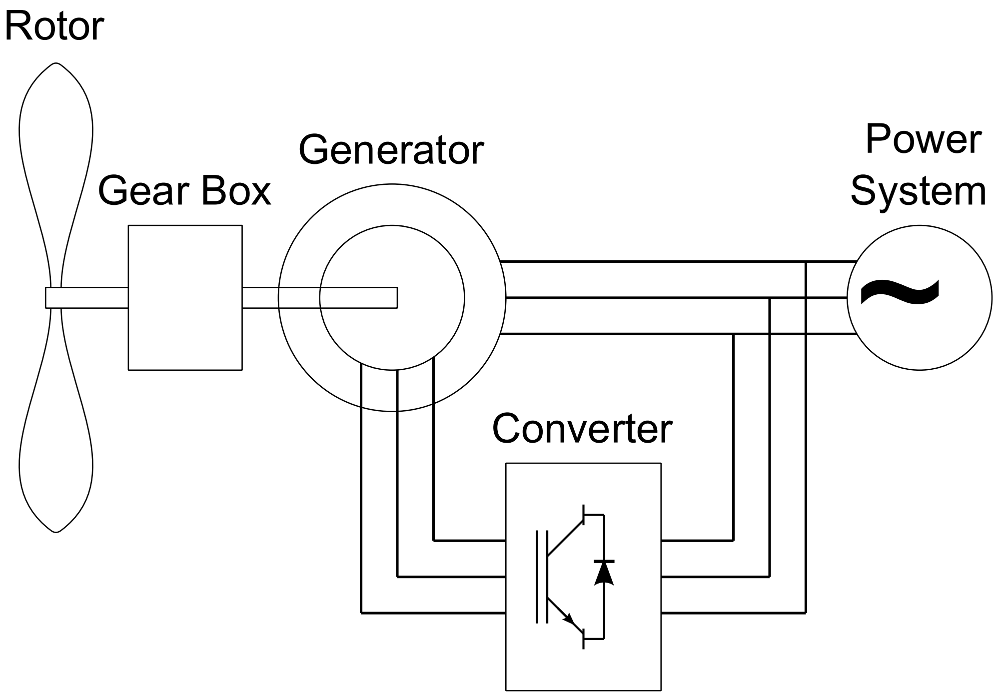
Modern windmills have evolved through several cycles of incredible innovation. Recently because of huge demand for energy created an interest in developing alternative energy especially on wind energy to generate electricity.
A wind turbine is a device that converts kinetic energy from the wind into electrical power. Wind turbines are fixed on a tower enables turbines to capture most of the wind energy. As the wind passes through the turbines it moves the blades which spin the shaft. The shaft is attached to a generator that produces electricity. Thus wind turbines convert the kinetic energy of the wind into mechanical energy. This mechanical energy can be converted into electricity by means of a generator.



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There are two types of wind turbines:
First one is horizontal-axis turbines and second one is vertical axis turbines.
In the horizontal-axis turbines the axis of rotation is horizontal. It is most widely used turbines. And It is more efficient than the vertical-axis windmills.

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Whereas in the vertical-axis design the axis of rotation is vertical. The blades will also be connected vertically.

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A group of wind turbines in the same location used to produce electricity they are called as wind farms.
Wind farms have advantages it makes easier to feed the produced electricity into the power grid.


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In recent years there is significant breakthroughs are happening in turbine technologies because of that the amount of energy produced by windmills has increased exponentially making wind power as an economically suitable energy source.
Wind power generation capacity in India has dominating in recent years.
In India the states of Tamil Nadu and Gujarat is the largest producer of wind energy according to the National Institute of Wind Energy (NIWE) reports.
India is the fifth-largest producer of wind power in the world.
Apart from India many countries also involved in the development of wind energy for example USA (California) Great Britain Greece Spain Netherlands and Denmark

Wind energy have many advantages compared to fossil fuels such as coal-based thermal energy.
Wind energy is a renewable and inexhaustible source of energy.
It is non-polluting it is low cost energy
It is a safe and clean source of energy and pollution-free energy resource
Wind energy is regarded as a potential energy source for solving todays energy problems.
Most importantly windmills do not produce any greenhouse gas emissions

Lest see the disadvantages of wind energy:
The main disadvantage of wind energy is it is unreliable energy resource. The degree wind availability and the speed of the wind are often uncertain.
Also high speed and large volumes of air is required to produce wind energy.
Wind does not blow at a constant rate and wind is not always consistent sometimes
it does not blow at all. As a result the wind energy must be harvested as and when it is available
The rotor noise produced by windmills causes noise pollution which is another disadvantage of wind energy
The windiest locations are often in remote locations thus it is location specific.

Just like with any other energy plant people oppose it because of aesthetic reasons that is due to visual pollution.
Winds mills have a costly setup
Sometimes windmills interferences to radio and TV signals
Furthermore the available evidence suggests that wind farms are harmful to birds birds often crash into the wind turbines. Many birds and bats die because of wind turbines.